

Princeton University
Intermediate Micro: Math Approach (ECO 310)
Spring 2025 Midterm Solutions—Andrea Wilson

Instructions: This is intended to be an 80-minute exam, and you have 100 minutes altogether (10 minutes for uploading your answers and 10 minutes of cushion). There are 6 multiple choice questions, each worth 4 points, which you can answer directly on Canvas. And three short answer questions, worth a total of 86. In the Short Answer section, REMEMBER TO SHOW YOUR WORK!, YOU'LL LOSE SIGNIFICANT CREDIT FOR ANSWERS THAT APPEAR BY MAGIC.

- please include the signed honor pledge in your short answer submission
- the actual total is 110, but I'll grade out of 100. So you can costlessly skip a 10-point question.

Multiple Choice Questions

1. A consumer's preferences over goods 1 and 2 are represented by $u(x_1, x_2) = x_1^{0.2}x_2^{0.4}$. Good 1 costs \$1 per unit, good 2 costs \$2 per unit, and income is \$15. How many units of good 2 should this consumer buy?

- (a) 3
- (b) 4
- (c) 5
- (d) None of these is correct

Answer: C. These are Cobb Douglas, with twice the exponent on good 2 as good 1 indicating that the consumer should spend twice as much on it, namely $\frac{2}{3}$ of income on good 2 and $\frac{1}{3}$ on good 1. So with \$15, he spends \$10 on good 2; at \$2 per unit, this buys 5 units.

2. On Monday, Homer Simpson buys a bundle of food and drink that costs \$90. On Tuesday, he buys a different bundle, for \$100, that would have cost \$80 at the Monday prices. In which of the following situations does he violate WARP:

- (a) If the Monday bundle would cost \$100 at the Tuesday prices.
- (b) If the Monday bundle would cost more than \$100 at the Tuesday prices.
- (c) Homer does not violate WARP.

Answer: A. On Monday, he buys a bundle for \$90 when the Tues one would have cost only \$80, revealing a strict preference for the Monday bundle. So he violates WARP if on Tuesday, he pays weakly more for the Tues bundle (indicating at least a weak preference); i.e. if the Monday bundle would have cost \$100 or less. This includes A but not B.

3. Assume utility $u(x_1, x_2) = \min\{4x_1, x_2\}$. The Hicksian demand bundle is given by:

- (a) $(4u^*, u^*)$
- (b) $(\frac{u^*}{4}, u^*)$

(c) $(u^*, \frac{u^*}{4})$

(d) $(u^*, 4u^*)$

Answer: B. To allocate money optimally, this consumer sets $4x_1 = x_2$ (so as to avoid wasting resources on extra units that yield no utility). Then plugging this into utility constraint $u(x_1, x_2) = u^*$, we get $4x_1 = x_2 = u^*$; solving yields the bundle in B.

4. There are two assets available: a risk-free asset, with a 10% return rate, and a risky asset, with a 20% expected return rate and variance $1/4$. If an individual divides his money between the two assets so that the standard deviation on his overall investment is $1/4$, then his overall expected return rate is:

(a) .15

(b) .1

(c) .2

(d) None of these is correct

Answer: A. With a fraction α in the risky asset, the consumer's overall standard deviation is α times the standard deviation $\sqrt{\frac{1}{4}} = \frac{1}{2}$ of the risky asset; so we're given $\frac{1}{2}\alpha = \frac{1}{4} \Rightarrow \alpha = \frac{1}{2}$. Thus the expected return rate, with half wealth at 10% and half at 20%, is 15% i.e. 0.15.

5. Jessica has utility $u(x_1, x_2) = x_1 + 2x_2$. She sets up a Lagrangian to find her best affordable bundle with income \$m. In which of the following cases do you expect the non negativity constraint on good 1 to be NON-binding?

(a) $p_1 = 2, p_2 = 5$

(b) $p_1 = 2, p_2 = 1$

(c) $p_1 = 2, p_2 = 3$

(d) More than one of these is correct.

Answer: A. Non-negativity on good 1 is NON-binding if the consumer wants a positive quantity of good 1. This happens with perfect substitute preferences whenever good 1 has more bang-per-buck, so if $\frac{MU_1}{MU_2} > \frac{p_1}{p_2} \Leftrightarrow \frac{1}{2} > \frac{p_1}{p_2} \Leftrightarrow p_2 > 2p_1$. This holds for A but neither B nor C.

6. Melsi has utility $u(B,T)=BT$ from B beef sticks and T toys. Initially, Melsi buys 16 beef sticks and 4 toys, when each beef stick costs \$1 and each toy costs \$4. What is the change in beef sticks due to substitution effects if the beef stick price rises to \$4? (no change in toy price).

(a) -16

(b) -8

(c) -4

(d) None of these is correct

Answer: B. These are Cobb-Douglas preferences with equal exponents, so to allocate money optimally between goods, Melsi sets $p_B B = p_T T$. At the new prices, this gives $4B = 4T \Leftrightarrow B = T$. Now, for the substitution effect, we first calculated the compensated

bundle that's optimized at new prices but yields old utility, so $B = T$ and $BT = 16 \cdot 4$; together, this gives $B^c = T^c = 8$. So finally, the substitution effect is $B^c - B^o = 8 - 16 = -8$.

Short Answer Questions

1. (32) Demand

Melsi's utility over beef sticks (B) and toys (T) is

$$u(B, T) = \left(B^{\frac{s-1}{s}} + T^{\frac{s-1}{s}} \right)^{\frac{s}{s-1}}$$

where s is a nonnegative parameter.

- (a) (10) Find Melsi's regular (Marshallian) demand functions, with p_B and p_T the beef stick and toy prices.

You should get

$$(B, T) = \left(\frac{m}{p_B^s (p_B^{1-s} + p_T^{1-s})}, \frac{m}{p_T^s (p_B^{1-s} + p_T^{1-s})} \right)$$

Answer: here, letting $X = \left(B^{\frac{s-1}{s}} + T^{\frac{s-1}{s}} \right)$, we have

$$MU_B = \frac{s}{s-1} X^{\frac{s}{s-1}-1} \frac{s-1}{s} B^{-\frac{1}{s}}, \quad MU_T = \frac{s}{s-1} X^{\frac{s}{s-1}-1} \frac{s-1}{s} T^{-\frac{1}{s}}$$

Dividing, everything cancels out except for the final term in each expression:

$$\frac{MU_B}{MU_T} = \frac{B^{-\frac{1}{s}}}{T^{-\frac{1}{s}}} = \left(\frac{T}{B} \right)^{\frac{1}{s}}$$

Since $s > 0$, this falls as $B \uparrow$ and $T \downarrow$, as desired for convexity. So we're looking for an interior solution, solving:

$$\text{equal bang-per-buck:} \quad \frac{MU_B}{MU_T} = \frac{p_B}{p_T} \Rightarrow \left(\frac{T}{B} \right)^{\frac{1}{s}} = \frac{p_B}{p_T} \quad (1)$$

$$\text{spend all money:} \quad p_B B + p_T T = m \quad (2)$$

Rearrange (1) as $\frac{T}{B} = \left(\frac{p_B}{p_T} \right)^s$, i.e. $T = \left(\frac{p_B}{p_T} \right)^s B$. Plug this into the budget line to get

$$p_B B + p_T \left(\frac{p_B}{p_T} \right)^s B = m \Rightarrow B = \frac{m}{p_B + p_T \left(\frac{p_B}{p_T} \right)^s} = \frac{m}{p_B^s (p_B^{1-s} + p_T^{1-s})}$$

And plugging this back into the optimal spending condition,

$$T = \left(\frac{p_B}{p_T} \right)^s B = \frac{m}{p_T^s (p_B^{1-s} + p_T^{1-s})}$$

- (b) (12) Show that for two of the options below (they're all true but pick the two you want to show), your demand functions in (a) approach those for:

- perfect complements as $s \rightarrow 0$
- perfect substitutes as $s \rightarrow \infty$
- Cobb-Douglas (with equal exponents) as $s \rightarrow 1$

Answer: as given in (a),

$$(B, T) = \left(\frac{m}{p_B^s (p_B^{1-s} + p_T^{1-s})}, \frac{m}{p_T^s (p_B^{1-s} + p_T^{1-s})} \right)$$

- as $s \rightarrow 0$, simply plug $s = 0$ into the demand function in (a) to get $B = \frac{m}{p_B^0 (p_B^1 + p_T^1)} = \frac{m}{p_B + p_T}$, and similarly $T = \frac{m}{p_B + p_T}$. This is what happens with perfect complements, where the optimal spending condition is $B = T$, and plugging this into budget line gives $B = T = \frac{m}{p_B + p_T}$
- as $s \rightarrow 1$, simply plugging $s = 1$ into the demand function in (a) yields $B = \frac{m}{p_B^1 (p_B^0 + p_T^0)} = \frac{m}{2p_B}$, and similarly $T = \frac{m}{2p_T}$. This is what the “magic rule” for Cobb Douglas with equal exponents says: spend half money on each good.
- as $s \rightarrow \infty$, demand for beef sticks from (a) is

$$\lim_{s \rightarrow \infty} \frac{m}{p_B^s (p_B^{1-s} + p_T^{1-s})} = \lim_{s \rightarrow \infty} \frac{m}{p_B \left(1 + \left(\frac{p_T}{p_B} \right)^{1-s} \right)} = \lim_{s \rightarrow \infty} \frac{m}{p_B \left(1 + \left(\frac{p_B}{p_T} \right)^{s-1} \right)}$$

As $s \rightarrow \infty$, $\left(\frac{p_B}{p_T} \right)^{s-1}$ tends to zero if $\frac{p_B}{p_T} < 1$, in which case this tends to $\frac{m}{p_B}$ (i.e. spend all money on B when it's cheaper, just as with perfect substitutes!) And to ∞ if $\frac{p_B}{p_T} > 1$, in which case this tends to zero with an exploding denominator. I.e. “buy zero units of B when it's more expensive”, again just like with perfect substitutes. And symmetrically for T .

Grading note: I generally gave 3 points for showing the limit clearly and 3 points for showing what the demand *should* be for perfect complements etc.

- (c) (10) Take as given that Melsi's expenditure function is $\frac{p_B u^*}{\left(1 + \left(\frac{p_B}{p_T} \right)^{s-1} \right)^{\frac{1}{s-1}}}$. Use this to

figure out how much money she would need to get utility $u^* = 1$ in the limit as $s \rightarrow 0$. Does your answer match your intuition for perfect complements? (Either explain why it does, or why you think there might be an error).

Answer: plugging in $u^* = 1$ and $s = 0$, this is

$$\frac{p_B}{\left(1 + \left(\frac{p_B}{p_T} \right)^{0-1} \right)^{\frac{1}{0-1}}} = \frac{p_B}{\left(1 + \frac{p_T}{p_B} \right)^{-1}} = p_B \left(1 + \frac{p_T}{p_B} \right) = p_B + p_T$$

This is intuitive: with perfect complements, it's optimal to set $B = T$. So to get utility $u^* = 1$, Melsi needs $B = T = 1$, i.e. one beef stick and one toy. This costs $p_B + p_T$.

2. (28) Expected Utility: Risk Aversion

(a) Consider the following bet:

$$\begin{cases} \text{win \$200} & \text{chance } \frac{1}{3} \\ \text{lose \$100} & \text{chance } \frac{2}{3} \end{cases}$$

i. (5) Explain (intuitively is fine) why a risk-averse decision-maker (DM) would reject this at ANY wealth level

Answer: the expected win with the gamble is $\frac{1}{3}(200) + \frac{2}{3}(-100) = 0$. So by taking the gamble, the DM expects to break even but with risk. Someone risk-averse would rather break even without risk, i.e. reject the gamble.

ii. (12) Next, explain why a risk-averse DM would find playing it twice even worse than playing it once. Specifically, complete the following proof:

- we know (1): $u(w_0 + 200) > \frac{1}{3}u(w_0 + 400) + \frac{2}{3}u(w_0 + 100)$ (why?)
- we know (2): $u(w_0 - 100) > \frac{1}{3}u(w_0 + 100) + \frac{2}{3}u(w_0 - 200)$ (why?)
- Now take a weighted sum of these inequalities ($\alpha(1) + (1 - \alpha)(2)$), and show that for suitably chosen α , you have answered the question

Answer: in (1), the LHS is the utility of gaining \$200 for sure, and the RHS is the expected utility of a risky gamble with expected wealth gain $\frac{1}{3}(400) + \frac{2}{3}(100) = 200$; the definition of risk-aversion is that for two gambles with the same expected wealth, the safe one is better. Likewise (2) holds because a risk-averse DM would rather lose \$100 for sure, than play a risky gamble with expected wealth loss $\frac{1}{3}(100) - \frac{2}{3}(200) = -100$.

Now, playing the gamble twice gives a $(\frac{1}{3})^2 = \frac{1}{9}$ chance of winning \$400 (win \$200 both times), a $(\frac{2}{3})^2$ chance of losing \$200 (lose \$100 both times), and a $\frac{4}{9}$ chance of winning \$200 and losing \$100 for net gain \$100. the EU of playing once is $\frac{1}{3}u(w_0 + 200) + \frac{2}{3}u(w_0 - 100)$. So we want to show

$$\frac{1}{3}u(w_0 + 200) + \frac{2}{3}u(w_0 - 100) > \frac{1}{9}u(w_0 + 400) + \frac{4}{9}u(w_0 - 200) + \frac{4}{9}u(w_0 + 100)$$

This is precisely what you get by adding $\frac{1}{3}(1) + \frac{2}{3}(2)$.

(b) (11) Now modify the bet in (a) to have the same prizes, but now equally likely. And assume $u(w) = \ln w$. Could it happen that a poorer decision-maker accepts the bet while a wealthier one rejects it? (Either find initial wealth levels where this would happen, or explain why you cannot).

Answer: the expected utility of the bet is now $\frac{1}{2} \ln(w_0 + 200) + \frac{1}{2} \ln(w_0 - 100)$. A DM with wealth w_0 will accept this whenever

$$\frac{1}{2} \ln(w_0 + 200) + \frac{1}{2} \ln(w_0 - 100) > \ln w_0$$

Subtracting half the RHS from each LHS term, and multiplying by 2, this rearranges to

$$\ln \frac{w_0 + 200}{w_0} + \ln \frac{w_0 - 100}{w_0} > 0 \Leftrightarrow \ln \left(\frac{w_0 + 200}{w_0} \frac{w_0 - 100}{w_0} \right) > 0$$

So a DM with wealth w_0 accepts whenever

$$\frac{w_0 + 200}{w_0} \frac{w_0 - 100}{w_0} > 1 \Leftrightarrow w_0^2 + 100w_0 - 20000 > w_0^2 \Leftrightarrow w_0 > 200$$

So no it could not happen: anyone with more than \$200 in initial wealth accepts, so if a poorer one takes it, so will a wealthier one.

3. (26) Expected Utility: Optimal Decisions

A risk-averse decision-maker with wealth w_0 would like to invest \$100 in total, some in “asset 1” and some in “asset 2”. The two assets are correlated in the following way (note, you can do this even if you have no idea what correlation is):

- in state A (chance $\frac{1}{3}$), return rates are $r_1^A > r_2^A$ for assets 1 and 2
- in state B (chance $\frac{1}{3}$), both assets have the same return rate, r^B
- in state C (chance $\frac{1}{3}$), return rates are $r_1^C < r_2^C$ for assets 1 and 2

- (a) (8) Write out the expected utility from investing $\$x$ in asset 1 and $\$100 - x$ in asset 2. Answer: in state A, the wealth gain is $r_1^A x + r_2^A(100 - x)$. In state B, the wealth gain is $100r^B$. And in state C, the wealth gain is $r_1^C x + r_2^C(100 - x)$. So

$$EU = \frac{1}{3}u(w_0 + r_1^A x + r_2^A(100 - x)) + \frac{1}{3}u(w_0 + 100r^B) + \frac{1}{3}u(w_0 + r_1^C x + r_2^C(100 - x))$$

- (b) (12) Show that in an interior solution, the optimal x satisfies the following equation, where w^A final wealth if state A happens and w^C is final wealth if state C happens. If you cannot get this, assume it anyway in part (c).

$$\text{interior solution: } \frac{u'(w^C)}{u'(w^A)} = \frac{r_1^A - r_2^A}{r_2^C - r_1^C} \quad (*)$$

Answer: the DM chooses x to maximize expected utility. In an interior solution, the FOC for the optimal x is

$$EU'(x) = 0$$

Differentiating the expression in (a),

$$\frac{1}{3}(r_1^A - r_2^A)u'(w^A) + \frac{1}{3}(r_1^C - r_2^C)u'(w^C) = 0$$

which rearranges to (*).

Most of you were asked about a corner solution: if the RHS of (*) is smaller than the LHS even at $x = 100$, it means

$$EU'(x) = (r_1^A - r_2^A)u'(w^A) - (r_2^C - r_1^C)u'(w^C) > 0$$

So even at $x = 100$, EU is increasing in x , meaning the DM would like to invest more in asset 1; since he has set \$100 as his max, he'll pick a corner solution with $x = 100$.

- (c) (6) Under what condition on the return rates does the DM prefer state C to state A, assuming he invests optimally as in (*)?

Answer: he prefers state C if $w^C > w^A$; by risk aversion, u' is declining so this holds iff $u'(w^C) < u'(w^A)$. By (*), this holds if

$$\frac{r_1^A - r_2^A}{r_2^C - r_1^C} < 1$$